Husayn El Sharif

Senior Data Scientist • Machine Learning Engineer

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Professional Summary

Ph.D. Senior Data Scientist with 15+ years of experience in geospatial analytics, industrial decision-support systems, and applied machine learning across research and engineering domains. Recent commercial work includes predictive modeling, MLOps pipelines, and LLM-powered AI systems.

Technical Skills

Programming: Python, SQL, R, MATLAB

Machine Learning: Scikit-Learn, TensorFlow, PyTorch, XGBoost, Deep Learning

Data Engineering & MLOps: NumPy, Pandas, PySpark, MLflow, Airflow, Streamlit, Docker, Git

AI Engineering: Prompt Engineering, Fine-Tuning, Agentic RAG, Claude Code, Codex

Cloud & Data Platforms: Google Cloud, AWS, Databricks, Snowflake, Hugging Face

Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Geospatial Analysis: ArcGIS, QGIS, GeoPandas, Google Earth Engine

Certifications

- Google Cloud Professional Machine Learning Engineer (2026)
- Databricks Certified Generative AI Engineer Associate (2026)

Professional Experience

Senior Data Scientist (Research Engineer) Aug 2012 – present

Georgia Institute of Technology, School of Civil & Environmental Engineering

Atlanta, GA

- Developed and deployed an end-to-end ML pipeline using Sentinel-2 satellite imagery and Google Earth Engine to generate weekly chlorophyll-a maps across 150 km² of Lake Lanier at 30 m resolution for the Gwinnett County Department of Water Resources; trained county technicians to use the system for near-real-time water quality monitoring and early warning of harmful algal blooms, reducing reliance on costly field sampling.
- Developed bias-correction and spatiotemporal downscaling algorithms for 16 climate models across two emissions scenarios, producing 4-km daily climate projections while preserving temperature-precipitation correlations. Integrated climate, soil, and crop data into ML models to predict crop yield and irrigation demand and quantify long-range agricultural risks across the ACF River Basin.
- Ported and modernized the SAC-SMA hydrologic model from R to Python using OpenAI Codex, adding flexible time steps, units, and calibration objectives. Achieved 100% model fidelity through rigorous testing, enabling high-performance computational workflows to assess projected changes in Georgia's water availability through the year 2100.
- Developed and evaluated an artificial neural network (ANN) to predict daily solar radiation from meteorological data, reducing RMSE by up to 9.7% versus a calibrated empirical model and supporting long-term agricultural forecasting.
- Oversaw $400K+ in annual federal and state research funding as Assistant Director of the Georgia Water Resources Institute, evaluating grant proposals for technical merit and practical impact, coordinating research awards, and ensuring timely technical and financial reporting to federal agencies.
- Mentored graduate researchers in data science, programming, and ML methods, supporting publications and conference presentations.

Senior Data Scientist (Part-Time) Sept 2025 – present

Demand Delta

Marietta, GA

- Developed an XGBoost lead-scoring system using weather, permit, valuation, and geospatial data, projecting a 12% revenue lift (~$450K per 600 leads) and improving precision-recall performance by 3.3x over a heuristic baseline. Added Platt calibration and SHAP for explainable per-lead insights.
- Built an explainable RAG platform to match project requirements with qualified engineering firms, indexing 7K+ public profiles into ~70K retrieval documents using PostgreSQL/pgvector, embeddings, hybrid search, reciprocal rank fusion, and cross-encoder reranking.
- Built an AI-powered lead-enrichment pipeline with Python, Google Gemini, and the Google Places API to surface high-value industrial prospects for an engineering consultancy, replacing manual prospecting with a scored outreach list and accelerating lead qualification.

Engineering Consultant (Part-Time) Aug 2022 – Aug 2025

4Earth, Inc.

Kennesaw, GA

- Built a TensorFlow-based decision algorithm combining neural networks and dynamic programming to select optimal membrane cleaning actions at 10-minute intervals, evaluating 74K+ system states to balance operational costs, equipment wear, and membrane performance.
- Developed and calibrated a data-driven decision support system for wastewater treatment operations, applying sensor time-series analysis and rule-based algorithms to automate membrane cleaning, detect anomalies, and handle edge cases across 32 operating scenarios.
- Supported AWS IoT integration for near-real-time monitoring of 24 operational KPIs at 60-second intervals, enabling data-driven insights into wastewater treatment performance and supporting the development of AI-powered operational systems.
- Developed physics-based and 3D modeling workflows and trained engineers in their adoption, improving design collaboration, reducing physical prototyping iterations, and shortening project delivery time by 20%.

Data Scientist (Graduate Research Assistant) May 2010 – Aug 2012

Florida Atlantic University, Department of Civil, Environmental, & Geomatics Engineering

Boca Raton, FL

- Built and validated ML models to reconstruct missing environmental data, benchmarking 71 methods across 33 years of daily observations from 15 weather stations to improve data accuracy and statistical consistency for hydrologic and climate analyses.

Education

Ph.D.	Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2019
M.Sc.	Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2016
B.Sc.	Civil Engineering	Florida Atlantic University	Boca Raton, FL	GPA: 3.81	Aug 2011